

# JUNHYUN PARK

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## EDUCATION

- Daegu Gyeongbuk Institute of Science and Technology (DGIST)** Feb 2024 – Feb 2026  
M.S., Robotics & Mechatronics Engineering  
Supervisor: Prof. Minho Hwang · GPA: 4.25/4.3
- Daegu Gyeongbuk Institute of Science and Technology (DGIST)** Feb 2020 – Feb 2024  
B.E., Computer Science and Electrical Engineering (Double Major)  
GPA: 4.07/4.3 · *Summa Cum Laude*, Valedictorian

## RESEARCH & PROFESSIONAL EXPERIENCE

- DEEPNOID**, AI Research Lab — Multimodal Team Feb 2026 – Present  
Researcher (Alternative Military Service) · Medical foundation models for CXR, CT, and MRI: vision-language modeling (VLM), self-supervised learning (SSL), and report generation (RG).
- DGIST**, Surgical Robotics and Robot Manipulation Lab Dec 2021 – Feb 2026  
Graduate / Undergraduate Researcher · Supervisor: Prof. Minho Hwang
- Harvard Medical School**, Lab of Medical Imaging and Computation Jul 2023 – Aug 2023  
Research Intern · Supervisors: Prof. Synho Do and Prof. Kyungsu Kim

## PUBLICATIONS

\* denotes equal contribution.

### Journal Publications

- [1] **SAM: Semi-Active Mechanism for Extensible Continuum Manipulator and Real-time Hysteresis Compensation Control Algorithm.**  
**J. Park\***, S. Jang\*, M. Park, H. Park, J. Yoon, M. Hwang.  
*International Journal of Medical Robotics and Computer Assisted Surgery*, 2024. (IF = 2.3, Q2)
- [2] **Hysteresis Compensation of Flexible Continuum Manipulator using RGBD Sensing and Temporal Convolutional Network.**  
**J. Park\***, S. Jang\*, H. Park, S. Bae, M. Hwang.  
*IEEE Robotics and Automation Letters (RA-L)*, vol. 9, no. 7, 2024. (IF = 5.3, Q1)

### International Conference Publications

- [1] **GLINT: Sparsely Gated Vision-Language Alignment for Fine-Grained Radiology Representations.**  
J. Park\*, S. Lee\*, **J. Park**, H. Yun, H. Kim, S. Jeong, H. Kang, B. Yoon, K. Choi.  
*Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- [2] **OFF-CLIP: Improving Normal Detection Confidence in Radiology CLIP with Simple Off-Diagonal Term Auto-Adjustment.**  
**J. Park\***, C. Moon\*, D. Lee, K. Kim, M. Hwang.  
*Medical Image Computing and Computer Assisted Intervention (MICCAI)*, 2025. (Acc. 30%, Early Accept — Top 9%)
- [3] **Vibration-Assisted Hysteresis Mitigation for Achieving High Compensation Efficiency.**  
M. Park\*, C. An\*, **J. Park\***, J. Kang, M. Hwang.  
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
- [4] **Optimizing Base Placement of Surgical Robot: Kinematics Data-Driven Approach by Analyzing Working Pattern.**  
J. Yoon\*, **J. Park\***, H. Park, H. Lee, S. Lee, M. Hwang.  
*IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.

- [5] **Integrating ChatGPT into Secure Hospital Networks: A Case Study on Improving Radiology Report Analysis.**  
K. Kim\*, **J. Park\***, S. Langerica, A. Alkhadrawi, S. Do.  
*Conference on Health, Inference, and Learning (CHIL)*, New York, Jun 2024. (Acc. 34%)
- [6] **Design and Kinematics Modeling of Flexible Continuum Manipulator for Endoscopic Surgery.**  
S. H. Jang\*, **J. Park\***, M. Hwang.  
*The 22nd International Conf. on Control, Automation and Systems (ICCAS)*, Nov 2022.

#### Preprints

- [1] **SD-GRPO: Verifiable Segment Decomposition for Long-Form Vision-Language Generation.**  
H. Kim, S. Lee, H. Yun, **J. Park**, J. Park.  
*arXiv preprint*, arXiv:2606.09871, 2026.
- [2] **SSTL: Self-Sensing Tendon Loop for Hysteresis Modeling and Compensation in Tendon-Sheath Mechanisms.**  
M. Park\*, **J. Park\***, I. Ullah, C. An, M. Hwang.  
*arXiv preprint*, arXiv:2605.16870, 2026.
- [3] **MarUco: A Markerless 6D Pose Estimation Framework for Closed-Loop Control of Surgical Continuum Manipulators.**  
**J. Park\***, C. An\*, M. Park, I. Ullah, S. Park, M. Hwang.  
*arXiv preprint*, arXiv:2602.16365, 2026.

#### PATENTS

- [1] **Method for Correcting Robot Control Signal and Electronic Device Therefor.**  
M. Hwang, **J. Park**, S. Jang, D. H. Kang, D. S. Kwon, D. S. Kim, C. S. Kim, H. Kim, T. De Sa Rosa, H. W. Baek, S. M. Seol, U. J. Yang, D. H. Lee.  
*WIPO (PCT) International Publication*, No. WO2025143812A1, Jul 2025.  
Appl. No. PCT/KR2024/021174 · Assignees: DGIST & ROEN Surgical Inc.  
Korean national patent registration in preparation.

#### CHALLENGES

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|--------------------------------------------------------------------------------------------------------------|----------|
| – <b>VLM3D 2026</b> , MICCAI — MRI Abnormality Classification [ <i>1st place</i> ]                           | Sep 2026 |
| – <b>VLM3D 2026</b> , MICCAI — CT Abnormality Classification [ <i>2nd place</i> ]                            | Sep 2026 |
| – <b>TREAT-MMTB 2026</b> , MICCAI — Task 2: TB Diagnosis [ <i>2nd place</i> ]                                | Aug 2026 |
| Challenge paper: <a href="https://openreview.net/forum?id=CZZCTepmMj">openreview.net/forum?id=CZZCTepmMj</a> |          |
| – <b>TREAT-MMTB 2026</b> , MICCAI — Task 1: Cavity Detection and Segmentation [ <i>3rd place</i> ]           | Aug 2026 |
| Challenge paper: <a href="https://openreview.net/forum?id=VMeKaliJwV">openreview.net/forum?id=VMeKaliJwV</a> |          |

#### HONORS & AWARDS

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| – Talent Award of Korea, Minister of Education<br>National award to 100 outstanding young talents selected annually.                                                                                    | Dec 2025            |
| – Best Researcher Award, Dept. of RME, DGIST [ <i>top 1%</i> ]<br>Selected as one of only 2 graduate students across all departments.                                                                   | Jan 2025            |
| – Excellence Poster Award, Dept. of RME, DGIST                                                                                                                                                          | Jan 2025            |
| – IPESK Next Generation Engineering Researcher                                                                                                                                                          | Jan 2025            |
| – Outstanding Paper Award [ <i>top 2.2%</i> ]<br>“Hysteresis Compensation of Endoscopic Flexible Continuum Manipulator using Deep Learning,” <i>The 18th Korea Robotics Society Annual Conference</i> . | Feb 2023            |
| – Korea Presidential Science Scholarship, President of Korea                                                                                                                                            | Jun 2022 – Feb 2024 |
| – DGIST Presidential Fellowship [ <i>top 1%</i> ], President of DGIST<br>Awarded to only 2 students from the same school year at DGIST.                                                                 | Apr 2021 – Feb 2024 |
| – Korean College Mathematics Competition — Silver Medal                                                                                                                                                 | Dec 2020            |

Awarded by the Korean Mathematical Society.

- Dean’s List 2020–2023  
Spring/Fall 2020, Spring 2021, Fall 2022, Spring/Fall 2023.

## FUNDED RESEARCH PROJECTS

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- **Vibration-Assisted Tendon-Sheath Friction Reduction** Aug 2025 – Feb 2026  
National Research Foundation of Korea (NRF), Seed Research Program (MSIT). · Researcher.
- **Body-Ownership-Based Wearable Augmentation Robot Technology (Phase 2)** 2022 – 2026  
Ministry of Science and ICT (DGIST Institutional R&D). · Researcher.
- **Intelligent Guidance System for Sleeve Gastrectomy (Bariatric Surgery)** 2023 – 2025  
Korea Medical Device Development Fund (KMDF; MSIT/MOTIE/MOHW); CHA University consortium with DGIST. · Researcher.
- **Flexible Surgical Robotic Arm for Insertion along Arbitrary Retroflexed Paths** 2022  
Ministry of Science and ICT (DGIST Institutional R&D); in collaboration with ROEN Surgical Inc. · Researcher.

## PROJECTS

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- **Exoskeleton Robot Control Using EMG Signals** Dec 2021 – Dec 2022  
DGIST Undergraduate Group Research Project. Group Leader · Supervisors: Prof. Minho Hwang, Prof. Ji-Woong Choi.
- **Highly Autonomous Vehicle (Level 4)** Mar 2022 – Oct 2022  
Computer Vision Developer · Supervisor: Prof. Gyengho Choi.
- **Startup: Personalized Nutrition Salad** Jun 2021 – Dec 2021  
CTO · Supported by the Ministry of SMEs and Startups.
- **Design and Fabrication of a Compact Electric Vehicle** May 2020 – Nov 2020  
Team Member · Supervisor: Prof. Sehoon Oh.

## PROFESSIONAL SERVICES

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- **Reviewer:** MICCAI (2025); IEEE RA-L, IEEE/RSJ IROS (2024).
- **Teaching Assistant:** Undergraduate Research Group Project (2025); Artificial Intelligence Basics (2024).

## MEDIA & OUTREACH

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- **Invited Panelist, *Conversations with Future AI Leaders*** (KBS) 2026  
A televised talk show on the future of artificial intelligence, featuring the Minister of Science and ICT together with approximately 12 invited participants.

## TECHNICAL SKILLS

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**Languages:** Python, C/C++.

**ML / DL Frameworks:** PyTorch, HuggingFace (Transformers, Accelerate), MONAI, TensorFlow.

**Methods:** Deep Learning, Computer Vision, Vision-Language Models (VLM), Large Language Models (LLM), Self-Supervised Learning (SSL), Medical Image Analysis.

**Robotics:** ROS/ROS2, Isaac Sim, SolidWorks, Continuum Manipulators, Collaborative Robot, Tendon-Driven Control.